

Market Share

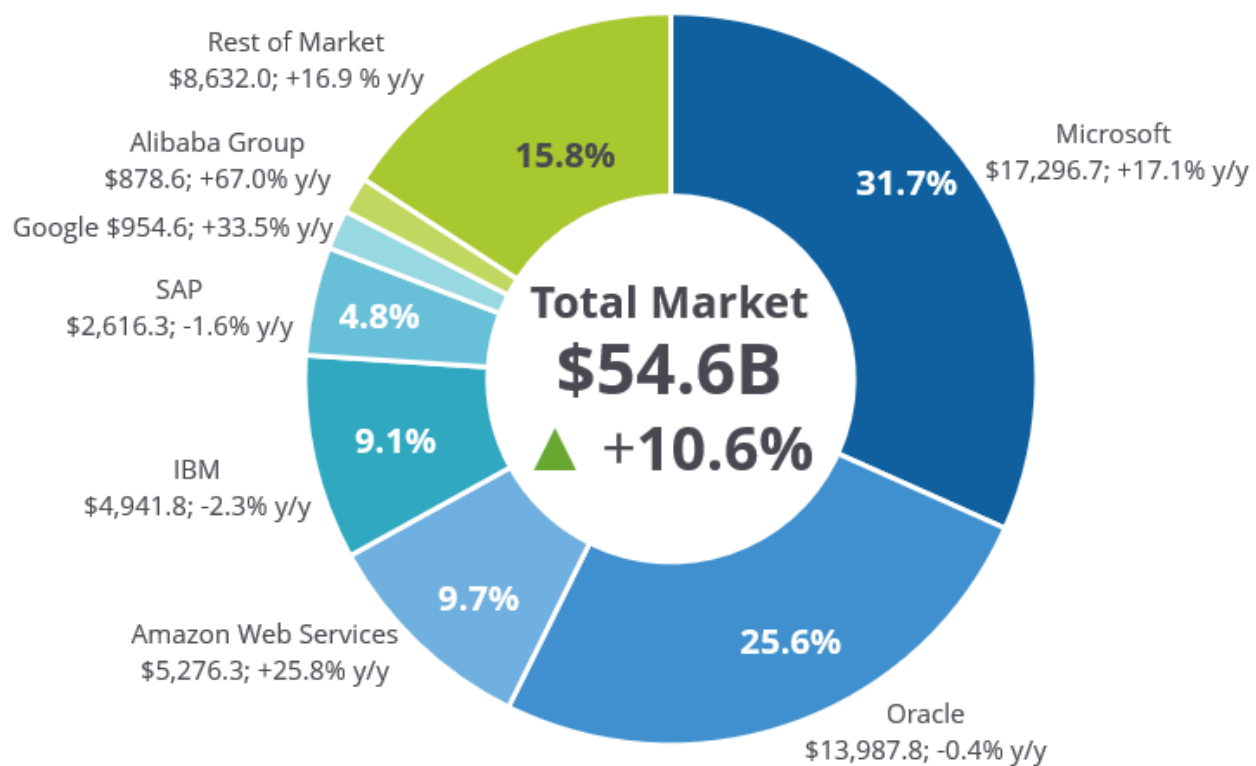
Worldwide Database Management Systems Software Market Shares, 2020: The Enterprise Journey to the Cloud

Carl W. Olofson

IDC MARKET SHARE FIGURE

FIGURE 1

Worldwide Database Management Systems Software 2020 Share Snapshot



Note: 2020 Share (%), Revenue (\$M), and Growth (%)

Source: IDC, 2021

EXECUTIVE SUMMARY

The worldwide market for database management systems (DBMSs) software grew by 10.6% from 2019 to 2020 (refer back to Figure 1). The two largest vendors, Microsoft and Oracle, account for over 57% of the market by revenue. Watchwords for this market seem to include speed, flexibility, agility, ease of use, consistency, durable persistence, and above all, security. Vendors showing the most substantial growth included Amazon, Google, Alibaba, and Snowflake. It should come as no surprise that the growth leaders included the major public cloud service providers. Many had expected that 2020 would be hampered significantly by the COVID-19 pandemic, but that effect was considerably less pronounced in the DBMS industry. In fact, it could even be said that the surprisingly rapid shift to remote work created a need for accessible data assets best served in the public cloud, thereby accelerating migration.

This IDC study contains revenue performance by vendor for the worldwide database management systems (DBMSs) software market, as well as vendor shares for each of the submarkets, which include data lake management systems, in-memory shared data managers, low-code DBMS, navigational DBMS, non-schematic DBMS, and relational DBMS. This data is reported for 2018-2020 and includes analysis of performance during that period by both current (as reported) currency and constant currency. It also includes vendor guidance, analysis of what shaped the market, including key vendors in 2020, and significant market events.

"The database management systems market is under pressure from multiple directions: cloud migration, IoT and other rapid streaming data sources, a rich variety of database models, the increasing desire to mix transactions and analytics, and an evolving approach to large-scale data collections, such as data lakes," said Carl Olofson, research vice president for Data Management Software research at IDC. "Vendors are reacting in various ways. Some are blending more functionality together, taking a multimodel approach to databases, others are promoting specialized single-model DBMSs. Some are combining data warehouse and data lake functionality with an overarching 'lake house' capability, while others are keeping them segregated."

ADVICE FOR TECHNOLOGY SUPPLIERS

The database management systems market is under pressure from multiple directions: cloud migration, IoT and other rapid streaming data sources, a rich variety of database models, the increasing desire to mix transactions and analytics, and an evolving approach to large-scale data collections, such as data lakes. Vendors are reacting in various ways. Some are blending more functionality together, taking a multimodel approach to databases, others are promoting specialized single-model DBMSs. Some are combining data warehouse and data lake functionality with an overarching "lake house" capability, while others are keeping them segregated. Watchwords for this market seem to include speed, flexibility, agility, ease of use, consistency, durable persistence, and above all, security. It seems clear that excelling in some of these means compromising on others. But no one can afford to compromise on security.

Given all these, the following should be considered by technology suppliers in this space:

- Public cloud service providers that include DBMSs services should be aware that users are increasingly interested in spreading database functionality across clouds. They may not wish to standardize on DBMSs that only operate on one cloud. Independent DBMS suppliers are aware of this and are exploiting it.

- Vendors committed to a single model of database should be all in on that model and accept that there are database workloads at which they excel and others that they do not adequately address. They should focus on what they do best and develop partnerships with others to provide their customers with a broader functional portfolio in which they are key players.
- Vendors that offer support for multiple database models in a single DBMS typically derive their sales opportunities from one of those models, which is primary. They need to understand that and aim their marketing at securing that market segment or risk losing ground, and possibly credibility, in competition with those focused more completely on the models that they consider secondary.
- Vendors that offer support for multiple database models, each in a specialized DBMS, must provide both integration and operational coordination capabilities that enable users to manage data across the suite as a common collection, rather than as disjoint databases that have no formal relationship with one another.
- All vendors in this market should accept that their competitors may not be content to "stay in their swim lanes" but broaden their scope, either through organic development or acquisition, to address new workloads. Analytic database vendors may venture into transactions or IoT and in-memory data managers may add transactions and query to their capabilities. This market is likely to be functionally fluid for some time to come.
- Right now, the "open source" attribute still carries some weight, especially in the developer community. As more databases are available in the public cloud as cloud services, this distinction is likely to grow less significant. Customers will be more interested in the nuts-and-bolts characteristics of price and performance.
- Watch out for graph databases. This could become the breakout technology that combines functions previously viewed as distinct and takes over existing workloads even as it enables new ones.

MARKET SHARE

Table 1 displays 2018-2020 worldwide revenue and 2020 growth and market share for the database management systems software market. Thanks in no small measure to the impact of Azure, Microsoft assumed the top spot in this market overall. Vendors showing the most substantial growth included Amazon, Google, Alibaba, and Snowflake. It should come as no surprise that the growth leaders included the major public cloud service providers.

TABLE 1

**Worldwide Database Management Systems Software Revenue by Vendor,
2018-2020 (\$M)**

	2018	2019	2020	2020 Share (%)	2019–2020 Growth (%)
Microsoft	12,558.0	14,770.0	17,296.7	31.7	17.1
Oracle	14,919.3	14,043.5	13,987.8	25.6	-0.4
Amazon Web Services	2,949.7	4,193.7	5,276.3	9.7	25.8
IBM	5,025.7	5,060.5	4,941.8	9.1	-2.3
SAP	2,657.2	2,658.7	2,616.3	4.8	-1.6
Google	456.7	714.9	954.6	1.7	33.5
Alibaba Group	278.9	526.2	878.6	1.6	67.0
Teradata	1,014.5	925.8	878.6	1.6	-5.1
Cloudera	609.0	561.5	656.2	1.2	16.9
InterSystems	517.0	516.9	544.4	1.0	5.3
Snowflake	114.7	236.9	518.4	0.9	118.8
MongoDB	181.2	277.1	439.9	0.8	58.7
Databricks	34.7	183.9	316.7	0.6	72.2
Tencent	90.1	164.8	260.7	0.5	58.2
Huawei	43.6	144.7	258.6	0.5	78.7
Fujitsu	242.9	248.1	249.8	0.5	0.7
Apple	208.5	200.7	220.7	0.4	9.9
Hewlett Packard Enterprise	195.1	202.9	210.6	0.4	3.8
DataStax	139.0	174.5	196.4	0.4	12.5
MarkLogic	138.1	149.3	152.1	0.3	1.8
Broadcom	129.6	129.7	133.4	0.2	2.8

TABLE 1**Worldwide Database Management Systems Software Revenue by Vendor,
2018-2020 (\$M)**

	2018	2019	2020	2020 Share (%)	2019–2020 Growth (%)
Salesforce.com	82.9	100.6	120.7	0.2	20.0
EnterpriseDB	81.3	83.0	114.1	0.2	37.5
Micro Focus	121.5	113.7	105.3	0.2	-7.4
Software AG	103.5	103.0	103.9	0.2	0.8
HCL Technologies	110.7	95.1	98.1	0.2	3.1
Neo4j	52.0	64.0	77.0	0.1	20.3
VMware	48.7	59.1	71.7	0.1	21.3
Empress Software	65.1	66.7	67.5	0.1	1.2
Hitachi	57.8	59.2	59.0	0.1	-0.3
Couchbase	40.6	46.3	54.5	0.1	17.7
China Telecom	4.1	21.5	45.1	0.1	109.2
GBase	32.5	33.3	44.4	0.1	33.2
Rocket Software	48.4	46.8	43.0	0.1	-8.0
Kingbase	17.8	21.4	37.8	0.1	76.5
Unisys	45.5	39.8	35.9	0.1	-9.9
TmaxSoft	25.9	29.9	34.6	0.1	15.7
China Mobile	–	–	29.5	0.1	NA
Dameng	10.6	15.0	27.9	0.1	86.1
SAS	30.0	27.2	24.1	0.0	-11.4
Kingsoft	7.5	12.8	20.8	0.0	62.5
Hazelcast	13.8	15.7	20.2	0.0	28.2
Alpha Software	22.2	21.3	20.0	0.0	-6.2

TABLE 1**Worldwide Database Management Systems Software Revenue by Vendor,
2018-2020 (\$M)**

	2018	2019	2020	2020 Share (%)	2019–2020 Growth (%)
Exasol	15.0	16.6	19.1	0.0	15.4
GridGain	18.0	14.0	16.7	0.0	19.9
UCloud	10.6	12.2	15.8	0.0	29.4
UNICOM Systems	14.5	14.7	14.6	0.0	-0.9
NEC	13.6	14.5	14.5	0.0	0.1
Cincom Systems	13.5	13.3	12.5	0.0	-5.8
Other	1,948.5	2,105.3	2,247.1	4.1	6.7
Total	45,559.8	49,350.5	54,584.1	100.0	10.6

Source: IDC's Worldwide Semiannual Software Tracker, April 2021

Table 2 shows the DBMS software market vendor shares by submarket. Microsoft, Amazon, and Google were the most diverse in terms of both the number of submarkets in which they participated and the distribution of revenue across those submarkets. IBM was also diverse, but its main revenue sources are relational and navigational DBMS.

TABLE 2

Worldwide Database Management Systems Software Revenue by Vendor and Functional Market, 2020 (\$M)

	Data Lake Management Systems	In-Memory Shared Data Managers	Low-Code DBMS	Navigational DBMS	Non-Schematic Database Management Systems	Relational Database Management Systems (RDBMS)	Total
Microsoft	1,404.1	–	1,184.4	–	2,115.4	12,592.8	17,296.7
Oracle	45.7	101.6	–	–	25.3	13,815.2	13,987.8
Amazon Web Services	1,106.2	462.8	–	–	1,538.5	2,168.7	5,276.3
IBM	56.8	32.2	–	560.8	66.0	4,226.1	4,941.8
SAP	6.6	–	–	–	11.4	2,598.4	2,616.3
Google	23.4	5.5	612.0	–	224.3	89.4	954.6
Alibaba Group	295.0	–	–	–	56.0	527.6	878.6
Teradata	–	–	–	–	–	878.6	878.6
Cloudera	656.2	–	–	–	–	–	656.2
InterSystems	–	–	–	544.4	–	–	544.4
Snowflake	–	–	–	–	–	518.4	518.4
MongoDB	–	–	–	–	439.9	–	439.9
Databricks	–	–	–	–	316.7	–	316.7
Tencent	–	–	–	–	86.3	174.4	260.7
Huawei	126.3	–	–	–	–	132.3	258.6
Fujitsu	–	–	–	17.7	–	232.1	249.8
Apple	–	–	220.7	–	–	–	220.7
Hewlett Packard Enterprise	156.4	–	–	–	–	54.2	210.6
DataStax	–	–	–	–	196.4	–	196.4

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Worldwide Database Management Systems Software Revenue by Vendor and Functional Market, 2020 (\$M)

	Data Lake Management Systems	In-Memory Shared Data Managers	Low-Code DBMS	Navigational DBMS	Non-Schematic Database Management Systems	Relational Database Management Systems (RDBMS)	Total
MarkLogic	–	–	–	–	152.1	–	152.1
Broadcom	–	–	27.5	105.9	–	–	133.4
Salesforce.com	–	–	–	–	–	120.7	120.7
EnterpriseDB	–	–	–	–	–	114.1	114.1
Micro Focus	–	–	–	–	–	105.3	105.3
Software AG	–	25.9	–	78.0	–	–	103.9
HCL Technologies	–	–	–	9.5	–	88.5	98.1
Neo4j	–	–	–	–	77.0	–	77.0
VMware	–	15.5	–	–	–	56.2	71.7
Empress Software	–	–	–	–	–	67.5	67.5
Hitachi	–	–	–	–	–	59.0	59.0
Couchbase	–	–	–	–	54.5	–	54.5
China Telecom	30.0	–	–	–	–	15.1	45.1
GBase	–	–	–	–	–	44.4	44.4
Rocket Software	–	–	43.0	–	–	–	43.0
Kingbase	–	–	–	–	–	37.8	37.8
Unisys	–	–	–	16.8	–	19.1	35.9
TmaxSoft	–	–	–	–	–	34.6	34.6
China Mobile	24.2	–	–	–	1.8	3.5	29.5
Dameng	–	–	–	–	–	27.9	27.9
SAS	–	–	–	–	–	24.1	24.1

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Worldwide Database Management Systems Software Revenue by Vendor and Functional Market, 2020 (\$M)

	Data Lake Management Systems	In-Memory Shared Data Managers	Low-Code DBMS	Navigational DBMS	Non-Schematic Database Management Systems	Relational Database Management Systems (RDBMS)	Total
Kingsoft	–	–	–	–	–	20.8	20.8
Hazelcast	–	20.2	–	–	–	–	20.2
Alpha Software	–	–	20.0	–	–	–	20.0
Exasol	–	–	–	–	–	19.1	19.1
GridGain	–	16.7	–	–	–	–	16.7
UCloud	–	–	–	–	2.2	13.6	15.8
UNICOM Systems	–	–	–	–	–	14.6	14.6
NEC	–	–	–	–	–	14.5	14.5
Cincom Systems	–	–	–	9.1	–	3.5	12.5
Other	200.9	43.8	107.4	89.9	250.1	1,554.9	2,247.1
Total	4,131.8	724.3	2,214.9	1,432.0	5,613.9	40,467.2	54,584.1

Source: IDC's Worldwide Semiannual Software Tracker, April 2021

Data Lake Management Systems Submarket Vendor Shares

The vendor shares for the data lake management systems submarket of the DBMS market are provided in Table 3. Microsoft and Amazon lead in this space, with Microsoft Azure Synapse (which is mostly a data warehouse platform but combines some data lake functionality) plus HDInsight and Amazon Web Services (AWS) EMR on top. Cloudera, which practically introduced this submarket from a technology perspective as the original purveyor of commercially supported Hadoop, is third in this market. The Chinese online service platforms Alibaba, Huawei, and China Mobile showed the strongest growth. With the introduction of data platforms that are not quite DBMSs and lake house functionality, the future of this market is somewhat uncertain.

TABLE 3**Worldwide Data Lake Management Systems Software Revenue by Vendor, 2018-2020 (\$M)**

	2018	2019	2020	2020 Share (%)	2019–2020 Growth (%)
Microsoft	610.8	971.1	1,404.1	34.0	44.6
Amazon Web Services	561.4	838.1	1,106.2	26.8	32.0
Cloudera	609.0	561.5	656.2	15.9	16.9
Alibaba Group	81.2	178.3	295.0	7.1	65.5
Hewlett Packard Enterprise	141.0	149.0	156.4	3.8	5.0
Huawei	37.9	78.6	126.3	3.1	60.6
IBM	37.9	52.6	56.8	1.4	7.9
Oracle	26.0	34.7	45.7	1.1	31.6
China Telecom	2.9	15.6	30.0	0.7	91.8
China Mobile	–	–	24.2	0.6	NA
Google	8.8	17.2	23.4	0.6	36.0
Other	132.4	165.9	207.5	5.0	25.0
Total	2,249.2	3,062.6	4,131.8	100.0	34.9

Source: IDC's Worldwide Semiannual Software Tracker, April 2021

In-Memory Shared Data Managers Submarket Vendor Shares

Shares of the in-memory shared data managers submarket of the DBMS market are shown in Table 4. These products are poised for dramatic growth as streaming data from IoT and other sources look to drive adoption of this technology for data ingest purposes, and as applications distributed across systems and running as microservices in containers become more predominant; such applications require immediate availability of the kind of shared transient data that in-memory shared data managers offer. Amazon dominated this submarket in 2020, and Oracle was in second place because of the significance of Oracle Coherence in Oracle Fusion middleware. Nonetheless, we can expect other vendors to grab more share in the coming years.

TABLE 4**Worldwide In-Memory Shared Data Managers Software Revenue by Vendor, 2018-2020 (\$M)**

	2018	2019	2020	2020 Share (%)	2019–2020 Growth (%)
Amazon Web Services	285.7	374.3	462.8	63.9	23.7
Oracle	99.5	100.6	101.6	14.0	1.0
IBM	29.1	31.8	32.2	4.4	1.2
Software AG	23.0	24.5	25.9	3.6	6.1
Hazelcast	13.8	15.7	20.2	2.8	28.2
GridGain	18.0	14.0	16.7	2.3	19.9
VMware	8.3	11.2	15.5	2.1	38.3
Other	36.6	42.3	49.4	6.8	16.6
Total	514.0	614.4	724.3	100.0	17.9

Source: IDC's Worldwide Semiannual Software Tracker, April 2021

Low-Code DBMS Submarket Vendor Shares

Table 5 shows the vendor shares for the low-code DBMS submarket of the DBMS market. Although Microsoft dominated this submarket in 2020 with Microsoft Access, that product is not growing, and others, such as Google Bigtable and Airtable, are gaining significance.

TABLE 5**Worldwide Low-Code DBMS Software Revenue by Vendor, 2018-2020 (\$M)**

	2018	2019	2020	2020 Share (%)	2019–2020 Growth (%)
Microsoft	1,331.5	1,300.1	1,184.4	53.5	-8.9
Google	274.9	450.1	612.0	27.6	36.0
Apple	208.5	200.7	220.7	10.0	9.9
Rocket Software	48.4	46.8	43.0	1.9	-8.0
Broadcom	26.7	26.7	27.5	1.2	2.8
Alpha Software	22.2	21.3	20.0	0.9	-6.2
Other	99.4	104.7	107.4	4.9	2.7
Total	2,011.6	2,150.5	2,214.9	100.0	3.0

Source: IDC's Worldwide Semiannual Software Tracker, April 2021

Navigational DBMS Submarket Vendor Shares

The vendor shares for the navigational DBMS submarket of the DBMS market are provided in Table 6. This submarket is dominated by legacy mainframe navigational DBMSs such as IBM's IMS, Broadcom's CA-IDMS and CA-DATACOM, and Software AG's Adabas. But the bright light here is InterSystems with IRIS, a DBMS that has both object-oriented and relational characteristics and is very strong both in the healthcare sector and in a broad range of applications that demand transaction and advanced query support for complex data objects that include images and sound, in addition to the usual text, numeric, and binary data types.

TABLE 6**Worldwide Navigational DBMS Software Revenue by Vendor, 2018-2020 (\$M)**

	2018	2019	2020	2020 Share (%)	2019–2020 Growth (%)
IBM	630.0	598.0	560.8	39.2	-6.2
InterSystems	517.0	516.9	544.4	38.0	5.3
Broadcom	102.9	103.0	105.9	7.4	2.8
Software AG	80.6	78.6	78.0	5.4	-0.8
Fujitsu	18.8	18.5	17.7	1.2	-4.3
Unisys	25.2	20.8	16.8	1.2	-19.3
Other	103.8	110.0	108.5	7.6	-1.4
Total	1,478.2	1,445.8	1,432.0	100.0	-1.0

Source: IDC's Worldwide Semiannual Software Tracker, April 2021

Non-Schematic DBMS Submarket Vendor Shares

Vendor shares of the non-schematic DBMS submarket of the DBMS market are shown in Table 7. This submarket showed the strongest growth in DBMS overall in 2020. These products support applications that demand more dynamic flexibility than a schematic database can afford and a broader range of data structures and types. Microsoft, with Azure Cosmos DB, and Amazon, with AWS DynamoDB and DocumentDB for MongoDB, led in this space by a wide margin. The largest vendor outside of the cloud platform providers in this submarket was MongoDB, which also featured the strongest growth of the top 10 in 2020. There is strong demand for this technology, and most of these vendors can expect strong growth in the years ahead.

TABLE 7**Worldwide Non-Schematic Database Management Systems Software Revenue by Vendor, 2018-2020 (\$M)**

	2018	2019	2020	2020 Share (%)	2019–2020 Growth (%)
Microsoft	918.7	1,461.8	2,115.4	37.7	44.7
Amazon Web Services	808.2	1,168.1	1,538.5	27.4	31.7
MongoDB	181.2	277.1	439.9	7.8	58.7
Databricks	34.7	183.9	316.7	5.6	72.2
Google	112.3	163.2	224.3	4.0	37.5
DataStax	139.0	174.5	196.4	3.5	12.5
MarkLogic	138.1	149.3	152.1	2.7	1.8
Tencent	23.5	52.8	86.3	1.5	63.5
Neo4j	52.0	64.0	77.0	1.4	20.3
IBM	50.3	63.7	66.0	1.2	3.6
Alibaba Group	24.6	38.1	56.0	1.0	46.9
Couchbase	40.6	46.3	54.5	1.0	17.7
Oracle	23.2	24.1	25.3	0.5	4.9
SAP	7.4	8.3	11.4	0.2	36.5
Other	160.2	204.6	254.1	4.5	24.2
Total	2,714.1	4,080.0	5,613.9	100.0	37.6

Source: IDC's Worldwide Semiannual Software Tracker, April 2021

Relational DBMS Submarket Vendor Shares

The vendor shares for the relational DBMS submarket of the DBMS market are shown in Table 8. Once again, the leading RDBMS software vendor in 2020 was Oracle Corp., mainly on the strength of Oracle Database, but with a growing contribution by the open source MySQL. Microsoft, with its combination of Microsoft SQL Server (including SQL Data Warehouse), Azure SQL Database, and Azure Synapse (the data warehouse part), gained substantial ground in this submarket. The fastest-growing vendor was Snowflake with its cloud data warehouse service. EnterpriseDB led among open

source RDBMS distributors with its support of PostgreSQL and its commercial package, EDB Postgres.

TABLE 8

Worldwide Relational Database Management Systems Software Revenue by Vendor, 2018-2020 (\$M)

Vendor	2018	2019	2020	2020 Share (%)	2019–2020 Growth (%)
Oracle	14,770.6	13,884.1	13,815.2	34.1	-0.5
Microsoft	9,697.0	11,036.9	12,592.8	31.1	14.1
IBM	4,278.4	4,314.4	4,226.1	10.4	-2.0
SAP	2,648.0	2,646.9	2,598.4	6.4	-1.8
Amazon Web Services	1,294.4	1,813.2	2,168.7	5.4	19.6
Teradata	1,014.5	925.8	878.6	2.2	-5.1
Alibaba Group	173.2	309.7	527.6	1.3	70.3
Snowflake	114.7	236.9	518.4	1.3	118.8
Fujitsu	224.0	229.6	232.1	0.6	1.1
Tencent	66.6	112.0	174.4	0.4	55.7
Huawei	5.7	66.1	132.3	0.3	100.1
Salesforce.com	82.9	100.6	120.7	0.3	20.0
EnterpriseDB	81.3	83.0	114.1	0.3	37.5
Micro Focus	121.5	113.7	105.3	0.3	-7.4
Google	58.6	80.4	89.4	0.2	11.2
HCL Technologies	98.2	85.3	88.5	0.2	3.8
Empress Software	65.1	66.7	67.5	0.2	1.2
Hitachi	57.8	59.2	59.0	0.1	-0.3
VMware	40.4	47.9	56.2	0.1	17.4
Hewlett Packard Enterprise	54.1	54.0	54.2	0.1	0.5
GBase	32.5	33.3	44.4	0.1	33.2

TABLE 8**Worldwide Relational Database Management Systems Software Revenue by Vendor, 2018-2020 (\$M)**

Vendor	2018	2019	2020	2020 Share (%)	2019–2020 Growth (%)
Kingbase	17.8	21.4	37.8	0.1	76.5
TmaxSoft	25.9	29.9	34.6	0.1	15.7
Dameng	10.6	15.0	27.9	0.1	86.1
SAS	30.0	27.2	24.1	0.1	-11.4
Kingsoft	7.5	12.8	20.8	0.1	62.5
Exasol	15.0	16.6	19.1	0.0	15.4
Unisys	20.3	19.0	19.1	0.0	0.5
China Telecom	1.2	5.9	15.1	0.0	155.6
UNICOM Systems	14.5	14.7	14.6	0.0	-0.9
NEC	13.6	14.5	14.5	0.0	0.1
UCloud	9.1	10.3	13.6	0.0	31.9
Other	1,447.6	1,510.4	1,561.9	3.9	3.4
Total	36,592.7	37,997.2	40,467.2	100.0	6.5

Source: IDC's Worldwide Semiannual Software Tracker, April 2021

WHO SHAPED THE YEAR

Five vendors that shaped the year 2020 are mentioned in the sections that follow.

Microsoft

Microsoft continues to attract its user base to the public cloud service Azure and to serve them with a variety of data management products, including the non-schematic Cosmos, the relational OLTP DBMS Azure SQL Database, and a product that combines data warehouse with data lake and lake house functionality: Azure Synapse. For classic data lake support, there's HDInsight. While there is still a substantial number of users with Windows on premises, it seems clear that they will, in time, evolve to the cloud, and Azure offers a wide variety of data management and analytics options. Microsoft also

offers database services for MySQL, PostgreSQL, MongoDB, Cassandra, and Databricks and a data caching service based on Redis.

Oracle

Oracle is deep in transition to a web services-based company, but for now, a substantial number of customers are either in a conventional on-premises mode or building hybrid solutions. For them, Oracle emphasizes Oracle Exadata as their transition platform, since they can run it either on premises or in Oracle Cloud Infrastructure (OCI) with the exact same characteristics. Oracle Autonomous Database is picking up some momentum, with an impressive combination of autonomic functionality and full service that ensures that users need not worry about release or patch management, resource management, or even database tuning. All of it is done for them. Oracle also recently added (in January 2021) a low-code cloud service called Oracle APEX Cloud Service. This is a fully managed APEX implementation (APEX is Oracle's low-code development tool, which is built into Oracle Database) in OCI that enables nontechnical users to set up a database instance, specify a simple database application, and run it without requiring any training or expertise. Oracle claims that initiatives like this, plus an aggressively affordable pricing plan, are attracting new, smaller customers and increasing options for the larger ones. In addition, Oracle has recently stepped up its support for MySQL with a scalable analytic transaction processing cloud configuration called HeatWave.

Amazon Web Services

AWS offers the leading public cloud service and fills it with all kinds of goodies for developers. In addition to the platform, AWS offers a range of database technologies directed at every need. AWS has taken the position that the best approach to data management is to use a DBMS that is purpose built for the task at hand. To this end, AWS offers Aurora for relational transaction support, Redshift for data warehouse support, DynamoDB (key value) or DocumentDB for MongoDB (document) for non-schematic database applications, Keyspaces in support of Cassandra implementations, EMR for data lake management, Neptune for graph databases, Timestream for time series implementation, ElastiCache for caching and session management (either Memcached or Redis versions as well as MemoryDB for Redis), and QLDB as a durable ledger database platform. These are linked together by a range of data integration services. The idea is that every application development team can work with just the right type of database for its workload. Since AWS attracts a large number of software-as-a-service (SaaS) vendors that require technology specific to their problem spaces, this approach has great appeal.

Snowflake

Snowflake has taken the DBMS market by storm with its rapid growth from the start. It seems the market was ready for a fully relational DBMS that exhibits characteristics of scaling on demand plus high-speed query performance for large databases. Fully supported as a cloud service on any of the three major cloud platforms, Snowflake has an optimized columnar implementation combined with a "micro-partitioning" scheme designed to ensure rapid, dynamic scaling without database reorganization. In addition to this, Snowflake has an unusual "data sharing" feature that enables users to make data in their database available for access by others, either within the organization or outside, including non-Snowflake users that may be their clients or partners. There is also provision for unstructured data storage and query through its data lake implementation. All of this is integrated into something that Snowflake calls Data Cloud. It seems clear from Snowflake's success that the move to abandon data warehouses in favor of supposedly cheaper analytic platforms has been turned around,

and many are seeking the reliable consistency of a relational database for enterprise-level data collection, management, and analysis.

MongoDB

MongoDB offers a document DBMS based on a free open source foundation, which it owns and manages. The "community" product can be downloaded and used without charge, but generally speaking, for production-grade implementations, it's best to subscribe to the full-featured version. MongoDB's early success was entirely on premises, but for the past couple of years, growth has been driven by cloud adoption, and MongoDB seems to be having great success with its managed cloud database service Atlas. Part of MongoDB's success can be tied to the company's large and very enthusiastic developer community, which not only uses the technology but promotes it with their employers and clients. Committed to the document model, MongoDB was the first document DBMS to come out with full ACID transaction support. MongoDB's product is a JSON document DBMS that offers fine scaling and operational performance, but only limited query capability, since it has no schema or overall mechanism for joining document data for analysis. There are some facilities available in this regard, however. Its most common implementations appear to be with edge data management, customer-facing applications (such as ecommerce), and gaming.

MARKET CONTEXT

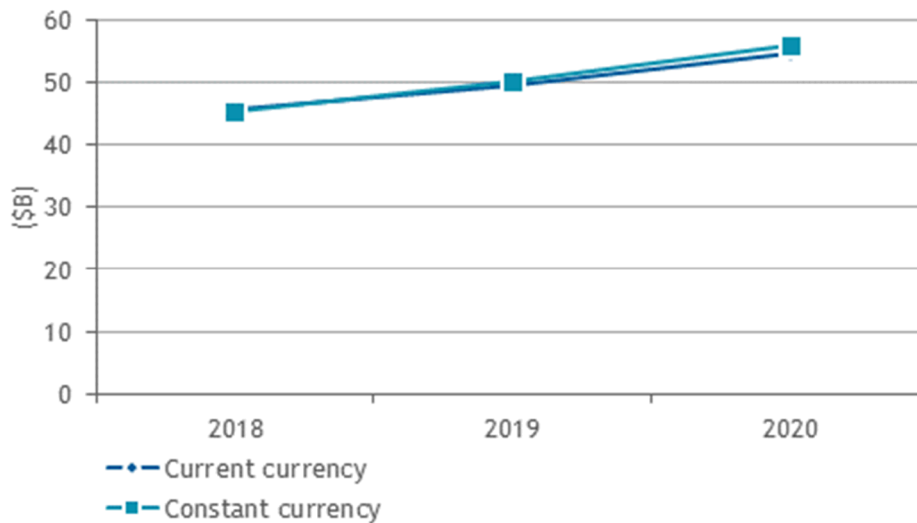
Many had expected that 2020 would be hampered significantly by the COVID-19 pandemic, but that effect was considerably less pronounced in the DBMS industry. In fact, it could even be said that the surprisingly rapid shift to remote work created a need for accessible data assets best served in the public cloud, thereby accelerating migration. It was already clear that critical data systems could not be reduced or curtailed during the crisis, so DBMSs serving mainline back-office applications were largely unaffected. The pandemic did cause some renewal of spending on the mainframe as enterprises that had hoped for a migration put off such plans because too much of the work would need to be done on premises.

In addition to the mixed, largely muted effects of the pandemic, general migration to the cloud, and the adoption of hybrid cloud models, led to increased investment in the relevant technology, including database systems, and to the adoption of various different database models including document and graph. So key drivers include the relentless move to the cloud and shift to cloud-native database technologies, increased adoption of business models that reach customers directly for such things as online ordering and "gig economy" reservations management (such as for app-driven car services and others), and the rising data volumes and need for more immediate analysis of streaming data. Inhibitors include continued dependency on old, complex datacenter configurations (particularly for large corporations) and, as always, the price-depressing effect of open source. As more and more open source is adopted as cloud services, however, the distinction between open source cloud services and proprietary cloud services is likely to diminish.

Figure 2 compares the constant and current currency performance of the market over the 2018-2020 period.

FIGURE 2

Worldwide Database Management Systems Software Revenue in Current and Constant Currency, 2018-2020



Source: IDC's Worldwide Semiannual Software Tracker, April 2021

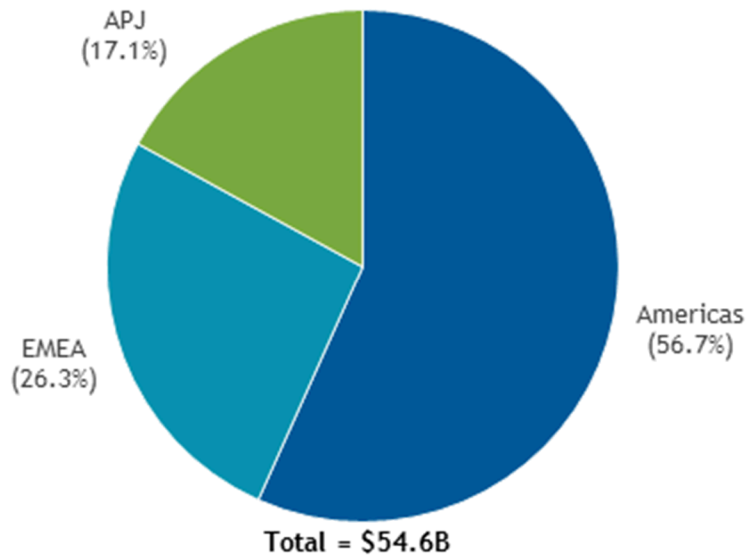
IDC includes constant currency measures in our market share documents to provide an indication of overall market developments excluding the effects of fluctuating exchange rates. The value of the U.S. dollar was up marginally against other currencies in 2020 over 2019. The 10.6% current currency or "as reported" growth in 2020 translates, when ignoring the impact of exchange rates, to constant currency growth of 11.1%. This is an important distinction as readers of this document digest its content.

As shown in Figure 3, the Americas remains the dominant region for this market with a 57% share.

Figure 4 shows that the larger portion of DBMS software is deployed on the user's premises, rather than in the public cloud.

FIGURE 3

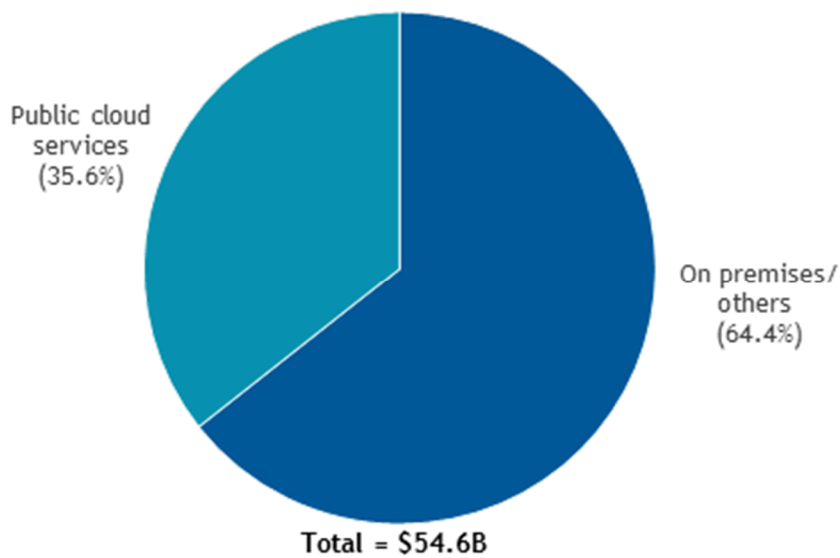
Worldwide Database Management Systems Software Revenue Share by Region, 2020



Source: IDC's Worldwide Semiannual Software Tracker, April 2021

FIGURE 4

Worldwide Database Management Systems Software Revenue Share by Deployment Type, 2020



Source: IDC's Worldwide Semiannual Software Tracker, April 2021

Significant Market Developments

The following represent significant DBMS market developments in 2020:

- Snowflake went public on September 16. Snowflake has enjoyed meteoric growth as enterprises seeking large-scale data analytics, and finding both data lakes and other data warehouse offerings wanting, turned to Snowflake with its dynamically scalable cloud-native architecture. That success is reflected in the fact that Snowflake was able to IPO much sooner than a software vendor normally could.
- Oracle introduced Oracle Exadata X8M Cloud Service on October 15. Oracle had released the Oracle Exadata X8M Database Machine in 2019, with its advanced use of the Intel Optane Persistent Memory Module and enhanced internal data movement using RoCE (RDMA over Converged Ethernet), but in 2020, this became available as a fully managed database service on Oracle Cloud Infrastructure.
- DataStax announced the availability of Astra as a native serverless cloud database service for Cassandra and related technologies on February 25. Given the profile of users that are attracted to Cassandra (which DataStax distributes and supports), this was a natural move.
- MongoDB announced Atlas Data Lake general availability on June 6. This was the latest in a series of developments (following the DataStax announcement) that see database management offered as cloud services. Several others have followed since. This development is significant because IDC sees database cloud services emerging as the dominant form of delivery for this technology going forward.
- Hewlett Packard Enterprise (HPE) introduced HPE Ezmeral, a data platform package including the technology of MapR, on June 23. Ezmeral is basically a data platform that combines data lake and lake house functionality to ease analytic application development and bring intelligence to users faster than before. It is part of a much larger data platform movement across major vendors. The data lake management software is based on MapR, a product based on the set of Apache Hadoop projects, but with significant proprietary enhancements.
- EnterpriseDB acquired 2ndQuadrant, the second-leading distributor of PostgreSQL, on September 30. This makes EnterpriseDB the dominant commercial distributor of PostgreSQL and an even more influential actor in the PostgreSQL community.
- MemSQL changed its name to SingleStore on October 27, signaling an emphasis on multipurpose relational functionality. The name MemSQL had always signaled an emphasis on the memory-centric nature of the DBMS, rather than other factors. The new name emphasizes the idea that this technology can be used for both high-performance transaction processing and analytic query support.

METHODOLOGY

The IDC software market sizing and forecasts are presented in terms of commercial software revenue. IDC uses the term *commercial software* to distinguish commercially available software from custom software. Commercial software is programs or codesets of any type commercially available through sale, lease, rental, or as a service. Commercial software revenue typically includes fees for initial and continued right-to-use commercial software licenses. These fees may include, as part of the license contract, access to product support and/or other services that are inseparable from the right-to-use license fee structure, or this support may be priced separately. Upgrades may be included in the

continuing right of use or may be priced separately. These are counted by IDC as commercial software revenue.

Commercial software revenue excludes service revenue derived from training, consulting, and systems integration that is separate (or unbundled) from the right-to-use license but does include the implicit value of software included in a service that offers software functionality by a different pricing scheme. It is the total commercial software revenue that is further allocated to markets, geographic areas, and sometimes operating environments.

Bottom-up/company-level data collection for calendar year 2020 began in January 2021 with in-depth vendor surveys and analysis to develop detailed 2020 company models by market, geographic region and, in some cases, operating environment.

The data presented in this document is IDC estimates only.

Note: All numbers in this document may not be exact due to rounding.

MARKET DEFINITION

Database Management Systems

A database management system (DBMS) is a software entity that manages a database in such a way that it may be queried and randomly updated. A database may be governed by a fixed, predefined structure to which the data must conform, called a schema, or not. This market includes six submarkets: relational DBMS, navigational DBMS, low-code DBMS (LCDBMS), NoSQL DBMS, data lake management systems, and in-memory shared data managers.

Some of the DBMSs in this market govern databases using schemas. The schema indicates the structure of the data, including its various data objects (which may be referred to as "record types," "object types," "classes," or "tables," depending on the paradigm of the DBMS in question and the types of relationships or associations among these objects). The schema contains rules governing the organization of such associations (such as optionality and cardinality) and the object formats (including field, column, or attribute types, valid values). It may also contain security rules governing access to the data. The schema enables multiple applications having separate code bases to share common data without sharing program source code.

Two submarkets within this market are for products that are entirely schematic: relational DBMS (RDBMS) and navigational DBMS (NDBMS). In the other submarkets, some databases also require schemas, some do not require them, but do support them, and still others have no concept of a schema at all. Low-code DBMSs are also usually schematic.

NoSQL (non-schematic) DBMSs, the fourth functional submarket, share some characteristics with schematic DBMSs. They support the common storage and retrieval of data that is optimized in a managed environment for quick saving and retrieval or for query or both. The key difference is that they do not require a schema and usually depend on program code or format elements embedded in the data, such as tags, to define their contents. They are used in cases where the data may not be well defined at the time of ingestion or where the application data requirements change so frequently that schematic management represents unacceptable overhead, slowing down the development process.

Relational Database Management Systems

The relational database management systems (RDBMS) submarket includes multiuser schematic DBMSs that are primarily organized according to the relational paradigm and that use SQL, or a protocol like SQL (such as ODBC or JDBC), as the foundational language for data definition and access. A relational DBMS includes a schema that defines the ways that data is accepted and returned in terms of tables having columns with rows uniquely identified by a primary key for each table and in terms of columns that are used to relate rows in different tables to each other through a value reference called a foreign key. Also included in this market are relational DBMSs that have been extended to support embedded tables or other nonrelational enhancements, include extended attribute types (such as graphical, geospatial, and audio) and object-oriented formalisms (such as data encapsulation), or directly support XML data. It also includes NewSQL DBMSs that may feature late schema binding, elastic scalability, and dynamic schema change.

Low-Code Database Management Systems

Low-code DBMSs are usually schematic, but the schema is simple, and its definition process is informal. LCDBMSs enable end users to define and manage their own data without any need for formal database administration (DBA). Such DBMSs typically reside on desktop operating environments, on workgroup servers, or in the cloud. They are used by knowledge workers either individually or in collaboration with others. These tools typically include a DBMS engine tightly integrated with a scripting language and report writer or another visualization facility, which provides a localized environment for data management and analysis.

Navigational Database Management Systems

A navigational DBMS is a schematic DBMS that is accessed using explicit navigation of the structure by the problem program, or which combines data in structures that cannot be broken out orthogonally and are optimized for rapid transaction processing. The data may be organized into fixed record structures, object class structures, or combinations include embedded structures.

Fixed Record Database Management Systems

A fixed record DBMS is typically organized using either the CODASYL or a proprietary structure. List oriented, hierarchical, B-tree indexed, network, and inverted lists are examples of organizations included in this category. The data is normally managed and accessed through a proprietary language or API, though most of these products also support access through standardized interfaces such as SQL, ODBC, JDBC, ADO.NET, Java Entity Beans, and various web services.

Object-Oriented Database Management Systems

Object-oriented DBMSs (OODBMSs) are designed to provide data storage and support services using an object-oriented architecture. An object-oriented DBMS supports the basic features of object-oriented development, including inheritance, polymorphism, encapsulation, and state. Unlike the other types of DBMSs, the OODBMS provides an implicit set of services associated with classes in the application object model that have been identified as "persistent," so there is no API, DML, or service call to be invoked by an application code. Because of the intimate relationship between the OODBMS data organization and the application, it is schematic, and its database schema must be aligned, either implicitly or explicitly, with the class structure of the application. Also an OODBMS must explicitly support (through header files and runtime libraries) each language to be used with it. These languages typically include Smalltalk, C++, C#, and Java.

Multivalue Database Management Systems

A multivalue DBMS is schematic and has an internal structure that supports nested data lists or structures. It layers over that structure a set of access and management services that operate according to one or more abstraction paradigms, which usually include both relational and object-oriented functionality. These products typically have proprietary DDL and DML and sets of APIs or drivers but can be accessed using standard SQL and other standard interfaces as well. Although they may support object-oriented data definition and manipulation, they differ from object-oriented DBMSs in that they make the database an explicit external service to which the application connects through an API or service layer. In contrast, object-oriented DBMSs enable the user to define database components as classes within the application object model and manipulate the database through methods inherited from a DBMS class library, thus making the DBMS itself an implicit service of the application's environment.

Non-Schematic Database Management Systems

A non-schematic DBMS can accept data without requiring that the structure and elements of the data be defined in advance. These include scalable data collection managers (the most common being Hadoop) and dynamic DBMSs. Because they do not require the use of SQL, dynamic DBMSs are sometimes called NoSQL database systems. There are two categories of non-schematic DBMS:

- **Semi-schematic DBMS:** The data may be governed by a schema, but one is not required. (Any data may be entered into the database that conforms to the general data format of the DBMS if no schema is present. If a schema is present, it governs the data and optimizes database operation on that basis.)
- **No schema DBMS:** No schema is required, and any data conforming to the general format of the database may be added.

The resulting collection of data may end up being rationalized under a schematic structure (in the semi-schematic case), mapped based on field names and values, or simply accessed by means of key-value pairs. The different types of non-schematic DBMSs are discussed in the sections that follow.

Document-Oriented Database Systems

Document-oriented database systems manage data blocks containing fields that are identified according to a generally accepted document format. The two most common such formats are XML and JavaScript Object Notation (JSON). Some document-oriented databases require the documents to be uniquely identified by a key value, and others do not. Some are semi-schematic, supporting the assertion of a document schema to ensure structural consistency, or simply facilitate document queries, and others are not. Some index all the fields, others allow users to designate some fields for indexing, and some don't index at all. Some document-oriented database systems have full DBMS capabilities for those that wish to take advantage of them, including schema support, optimized SQL query support, and relationship integrity constraint support.

Key-Accessible Database Systems

Key-accessible databases store data in a way that supports random retrieval by key value or retrieval in key-value order. There are two recognized key-accessible database system types:

- **Keyed block stores, also called key-value stores:** These are the simplest in concept. They store blocks of data that are identified by key value; the rest of the data is program defined. They incorporate no intrinsic knowledge of fields or relationships.

- **Keyed column stores, sometimes called wide column stores:** These store blocks of data-containing columns, which are lists of values. One of these columns is the key column. The columns are grouped into column families. Within each family, the values in each column are related to the values in every other column according to their position in the list. They are sometimes called wide column stores because column families often have very large numbers of columns, and the data is typically completely unnormalized.

Graph Database Management Systems

Graph DBMS software manages data as graph structures. These contain objects sometimes called "nodes" or "vertices" with recursive attributed relationships, sometimes called "edges." The attributes of the objects and relationships are called "properties." Unlike a fully schematic database, the structure of a graph database is derived from the relationship structure that is found in the instance data.

Such databases can be thought of as schematic in some cases because some graph DBMSs support the assignment of attributes and constraints for nodes and vertices in metadata. In a larger sense, however, schemas do not define data organization; instead, the database is designed to discover that organization. In some cases, the schema structure (i.e., the acceptable patterns of combinations of nodes, edges, and properties) may be predefined and fixed, and in other cases, it may be inferred from the data as it is ingested. Graph databases are used to capture and analyze extremely complex relationship instance structures. Common use cases include genealogy-based, spatial-based, and law enforcement (or intelligence service)-based "known associates" analysis. Graphs are also used in pharmacological research, epidemiology, and utility network analysis.

In-Memory Shared Data Managers

In-memory shared data managers (IMSDMs) manage data that is being used and constantly altered by running application processes. This software enables those processes, running either on a single system or multiple systems in a network, to share this data. The data is usually transient rather than recorded and so is relevant to application tasks at the specific time it is accessed and is generally contextually specific. As such, this software does not include database management systems, which manage recorded data that is application neutral and independent of context.

An IMSDM may have metadata that defines the data it manages, and it may also ensure recoverability of long application tasks through a persistent log or by redundant systems synchronization. An IMSDM may be associated with a back-end DBMS that records the final state of data collections that the IMSDM manages; that back-end DBMS may be specific to the IMSDM, or it may be a general-purpose DBMS, such as a RDBMS. Increasingly, IMSDM products have developed storage and query capabilities that enable them to act as managers of databases of record, and so the distinction between them and other categories of non-schematic DBMS is becoming obscure.

The IMSDM is typically memory based, managing its data as collections in main memory (which may be swapped to disk in cases where the amount of memory is insufficient to handle all the data; in this way, it resembles a virtual memory manager). It may or may not use locking to control concurrent access, and it may or may not have a notion of write ownership of specific data elements. Even in such cases, its method of granting access to data is generally less formal than that of a DBMS.

The purpose of the IMSDM is to allow an application's processes to be spread across many servers in a network or nodes in a cluster while closely sharing data. Typical application architectures for IMSDM

include a service-oriented architecture (SOA) and a distributed process web application running on a cluster. Sometimes, an IMSDM is persistent, writing its data to disk for recoverability, and its vendor may call it a "database." It is not a DBMS, however, if it cannot be used to share data across applications (except by sharing program code) or query end users. Thus key-value stores used to share data across instances of an application for performance and scalability, without governing metadata for application-neutral data sharing, are also IMSDMs.

Data Lake Management Systems

Data lake management systems provide an environment for data management and usually are accompanied by tools that may be used to define, query, and secure the data they collect. They are optimized for large volume data collection and analytic processing. These systems are sets of cluster management, file management, and application execution software designed to facilitate applications that collect large amounts of data, sift through and organize that data, and perform analytics on the data. The data collection may or may not end up containing one or more databases.

The most common form of these products is that based on the Apache Hadoop family of open source software projects, though others are emerging based on collecting data in scalable storage (such as in the cloud) and accessing it through analytic components such as Apache Spark (which is also used in conjunction with Hadoop) and Presto. Revenue from support for the full family of Hadoop projects, including HBase and Hive, is counted in this market. HBase is a keyed column store, and Hive is used to define Hadoop data for SQL access.

RELATED RESEARCH

- *IDC TechBrief: Graph Databases* (IDC #US46433020, June 2021)
- *IDC's Worldwide Software Taxonomy, 2021* (IDC #US47588620, April 2021)
- *Experience in Migrating Databases to the Cloud* (IDC #US46433821, March 2021)
- *Significant Data Management Announcements in January 2021* (IDC #US45958920, February 2021)
- *Multimodel Versus Fit-for-Purpose DBMS: How to Choose* (IDC #US46432921, December 2020)
- *IDC FutureScape: Worldwide Data and Analytics 2021 Predictions* (IDC #US46920420, October 2020)
- *Worldwide Database Management Systems Software Forecast, 2020-2024* (IDC #US46556220, June 2020)
- *Worldwide Database Management Systems Software Market Shares, 2019: Managing Data Across the Spectrum* (IDC #US46556320, June 2020)

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